

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

November 2013

EC 401 Data Communication & Networking

Date: 12.11.2013, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a) What do you mean by Primitives? explain basic four classes of service primitives [02]
(b) Explain TCP/IP reference model. [05]
- Q - 2 (a) Compare LAN, MAN and WAN in detail. [07]
(b) Explain the following terms in brief. [04]
(i) protocol (ii) multicasting (iii) routers (iv) half-duplex communication
(c) Compare connection oriented and connection less services. [03]

OR

- Q - 2 (a) Explain connection establishment in TCP using three way handshaking. [07]
(b) Explain TCP segment Header in detail [07]
- Q - 3 (a) Explain structure of telephone system in detail. [07]
(b) Define switching. Compare circuit switching, message switching and packet switching in detail. [07]

OR

- Q - 3 (a) Explain shortest path routing algorithm in detail. [07]
(b) Draw and explain IP datagram format. [07]

SECTION – II

- Q - 4 (a) Explain twisted pair cable in brief. [02]
(b) Write a short note on electronic mail (email) [05]
- Q - 5 (a) Explain congestion control in datagram subnets. [07]
(b) Briefly discuss SIP and H.323 protocol in case of VoIP. [07]

OR

- Q - 5 (a) Explain World Wide Web (WWW) in detail. [07]
(c) Explain carrier sense multiple access (CSMA) protocols in detail [07]
- Q - 6 (a) Explain transposition ciphers and one time pads. [07]
(b) Explain bit stuffing and bit de-stuffing in brief. [03]
(b) What do you mean by Fragmentation? Explain in very brief. [04]

OR

- Q - 6 (a) Explain DNS (domain name systems) in brief. [05]
(b) Explain RSA algorithm in detail. [05]
(c) Compare X.25 and frame relay. [04]
